

Predicting the next viral slang

Erdos institute

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The pitch

Fox 5 Gen Alpha Slang

"Meltmaxxing"

"Chopped" "Clanker"

"67"

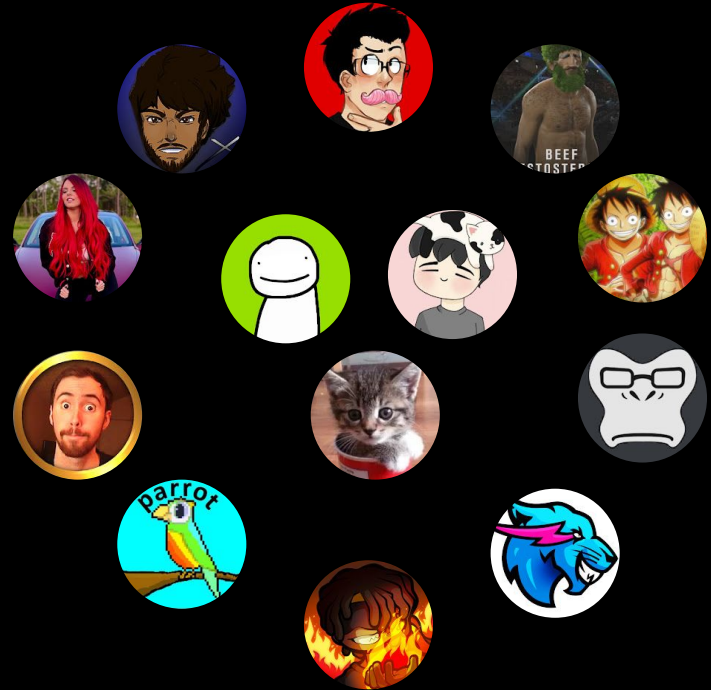
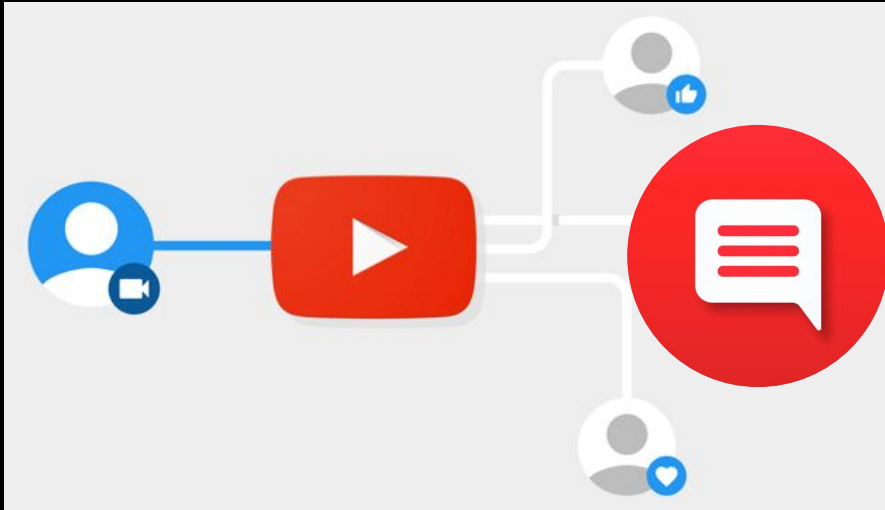
"SDIYBT" "Crashout"

Predicting virality helps take advantage of untapped potential



The data

Youtube Data API v3



Data curation

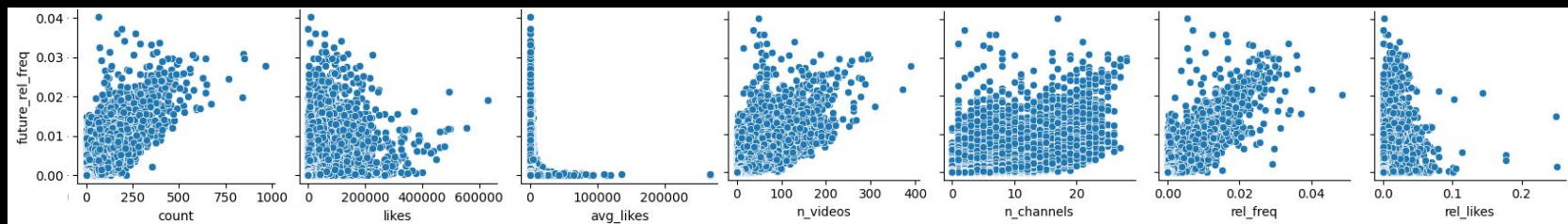
- Words which appeared exceptionally rarely (at most 4 times in all weeks) were removed from the dataset
- Exceptionally common words like "a" or "the" were also removed
- # words before filtering: 104,102
- # words after filtering: 11,156
- As a consequence of how the data was collected, the dataset includes much more comments from recent weeks compared to earlier weeks, so the models were trained on the most recent 20 weeks.

	video_id	text	likes	published_at	channel
0	BSv6hna2Ycs	3:22 zenin clan treatment	49	2026-06-15 11:44:27+00:00	1STUD
1	BSv6hna2Ycs	your goat stud i wacht you 4 years	29	2026-06-15 11:03:58+00:00	1STUD
2	BSv6hna2Ycs	I WATCHED YA FOR 2 YEARS!!!	6	2026-06-15 12:03:19+00:00	1STUD
3	BSv6hna2Ycs	2:21 Bro Noob Army Being Like God Slayer 🤖	3	2026-06-15 14:31:33+00:00	1STUD
4	BSv6hna2Ycs	Yo stud even though you probably not going to ...	3	2026-06-15 14:24:59+00:00	1STUD
...	...	...	...	...	...
39469	fb_W4pk5F5k	Insurance is to insure that the damages you do...	0	2025-05-25 06:08:22+00:00	xQcOW
39470	fb_W4pk5F5k	Why'd that restarted chatter say that was Dubstep	0	2025-10-23 02:38:50+00:00	xQcOW
39471	fb_W4pk5F5k	I love this bird, but not understanding the ne...	0	2025-05-26 03:55:53+00:00	xQcOW
39472	fb_W4pk5F5k	Bros yapping ruining a good video like alway	0	2025-05-24 18:50:43+00:00	xQcOW
39473	fb_W4pk5F5k	again the cam drivers often so overly exagger...	0	2026-02-04 06:16:46+00:00	xQcOW

Raw comment data

	word	time_id	count	likes	avg_likes	n_videos	n_channels	comments	total_likes	avg_comment_likes	rel_freq	rel_likes
0	0	1	28	6300	225.000000	24	12	9134	2886048	315.967594	0.003065	0.002183
1	0	2	42	32309	769.261905	36	15	12899	5234768	405.827428	0.003256	0.006172
2	0	3	30	7566	252.200000	22	9	11795	6134317	520.077745	0.002543	0.001233
3	0	4	24	4109	171.208333	21	8	11962	6002463	501.794265	0.002006	0.000685
4	0	5	34	9529	280.264706	30	12	13483	5755783	426.891864	0.002522	0.001656

Data filtered into statistics for each word for each week in the dataset.



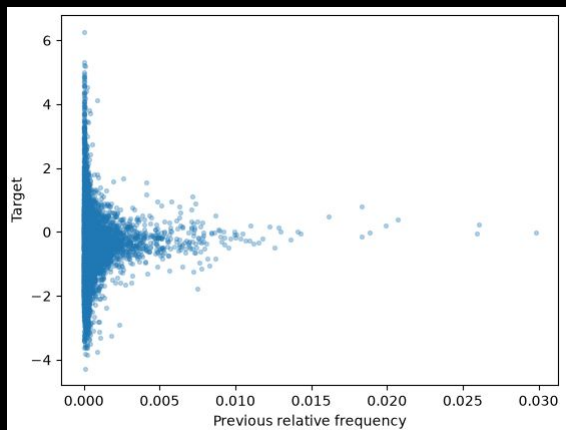
Plots comparing the statistics of a word for one week to the word's relative frequency the next week

Target

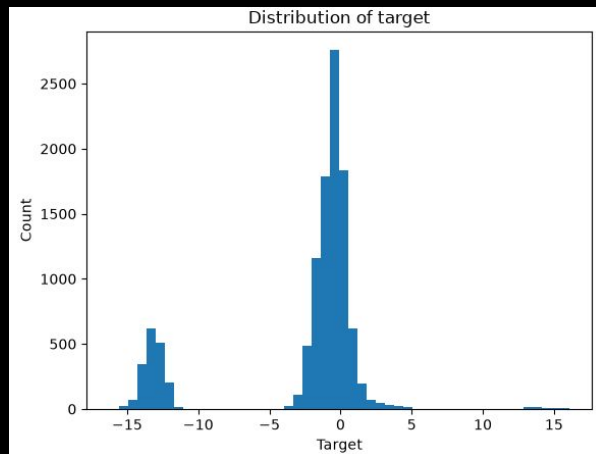
The target was defined to be

$$y = \log\left(\frac{\text{avg. relative frequency for final 3 weeks}}{\text{avg. relative frequency for previous 17 weeks}}\right)$$

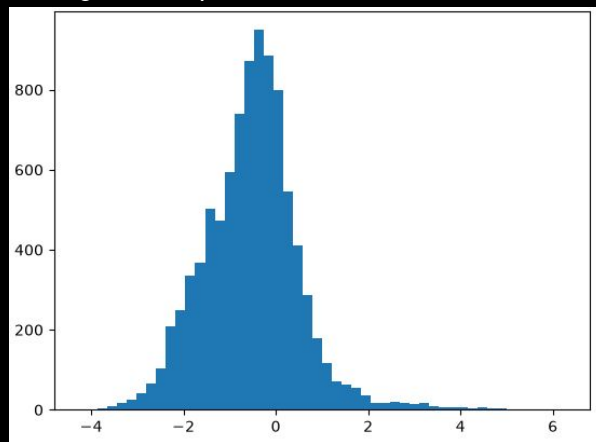
Outliers were found corresponding to words that appeared no times either in the final 3 weeks or the preceding 17 weeks. These were removed from the dataset before training models



Plot comparing the target to the relative frequency in the previous week (after outlier removal)



Histogram of y before outlier removal



Histogram of y after outlier removal

Feature Selection

The following weekly features were collected across the first 17 weeks:

1. Number of uses
2. Number of likes received by comments containing word
3. Average likes of comments containing word
4. Number of videos containing comments with word
5. Number of channels containing comments with word
6. Relative frequency
7. Likes per comment
8. Log growth of the above

Features were also aggregated across the first 17 weeks including:

1. Total number of weeks of use
2. Average count across weeks
3. Standard deviation of average count

Comments and future refinements:

- Relied on domain knowledge
- High likelihood of linear dependence likely to impact stability of linear models
- Likely to benefit from additional feature analysis (SHAP, regularization, etc.) to remove redundancy

Model Comparison

Models compared after hyperparameter tuning using RandomizedSearchCV within 5-fold nested cross-validation:

- Dummy Mean Regressor (baseline)
- Linear Regression
- Random Forest
- Extreme Gradient Boosting (XGBoost)
- Histogram-based Gradient Boosting (HistGB)

Model	RMSE	R2	Pearson	Spearman
Dummy mean	1.0501	-0.0003	-	-
Linear	0.9722	0.1429	0.3885	0.3258
Random Forest	0.8636	0.3228	0.5693	0.4065
XGBoost	0.8595	0.3294	0.5747	0.4149
HistGB	0.8641	0.3221	0.5687	0.4043

XGBoost improves nested CV RMSE by 18.15% over the dummy mean baseline (1.0501 → 0.8595).

Final Model: XGBoost

Selected using nested cross-validation, then retrained on the full training set and evaluated once on the held-out test set

Best Hyperparameters:

489 trees, max_depth = 6, min_child_weight = 7,
learning rate = 0.0091, subsample= 0.7839, colsample = 0.79,
reg_alpha = 0.0002, reg_lambda = 1.7249.

Held-out Test set performance

The tuned XGBoost generalizes well: the held-out RMSE (0.842) is consistent with the nested CV estimate (0.859 ± 0.020), indicating little evidence of overfitting. It also improves RMSE by 17% over the dummy baseline and achieves an R^2 of 0.31.

Model	RMSE	R^2	Pearson	Spearman
XGBoost	0.8423	0.3094	0.5568	0.4032
Dummy Mean	1.014	-0.0009	-	-

Top K	Overlap
Top 10	7/10
Top 50	28/50

Top-K retrieval: Number of truly viral words recovered among the model's highest-ranked predictions.

Classification Layer: Interpreting Virality

A word is defined **viral** if it's future relative frequency at least doubles, i.e., $y > \log(2)$

Threshold Selection (Cross validated out-of-fold training predictions)

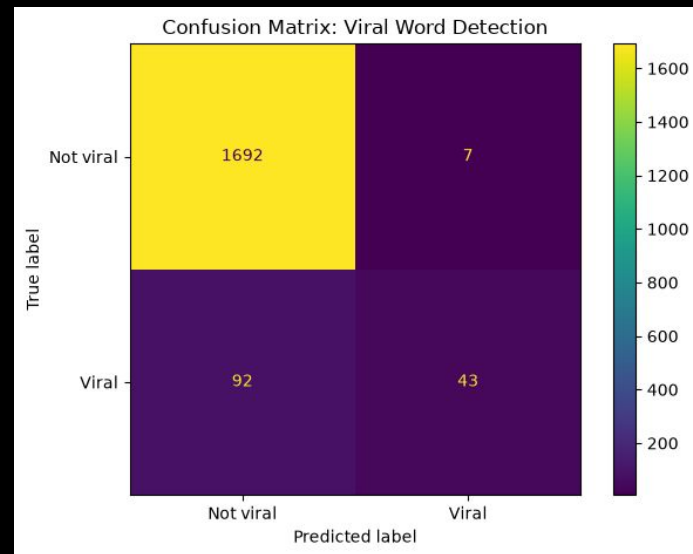
- Compute Precision-Recall curve using out-of-fold training predictions.
- Evaluate all possible prediction thresholds.
- Keep only positive thresholds with precision ≥ 0.80 .
- Choose the threshold with highest Recall (breaking ties by F1 score).

Selected threshold: $\hat{y} > 0.657$ (Precision = 0.80, Recall = 0.29 on the training set)

Statistics on Test Set

True Viral threshold	Prediction threshold	Precision	Recall	F1 Score	ROC-AUC	Average Precision
$y > \log(2)$	$\hat{y} > 0.657$	0.86	0.319	0.465	0.771	0.485

Test set: 1834 words · 135 truly viral · 50 predicted viral



Model's Top Predictions



Rasengan



Ky



KH4



Bolin



Ashe



Galio



Spurs



Sanji



Kaka



Zlatan

Future Work

- **Broader data collection:** Expand to Reddit, TikTok, and sources covering news, politics, gaming, and entertainment.
- **Improve text processing:** Merge spelling variants and treat common slang phrases such as "no cap" and "let him cook" as single units.
- **Rolling weekly forecasting:** Use features from week t to predict each word's growth from week t to week $t + 1$.
- **Viral-focused training:** Assign greater weights to words with large future growth so that missing a viral word is penalized more heavily than making a small error on an ordinary word.
- **External signals:** Include Google Trends, news activity, trailer releases, and major events to capture sudden viral spikes.